

EXPERIMENT - 4

Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Question 1

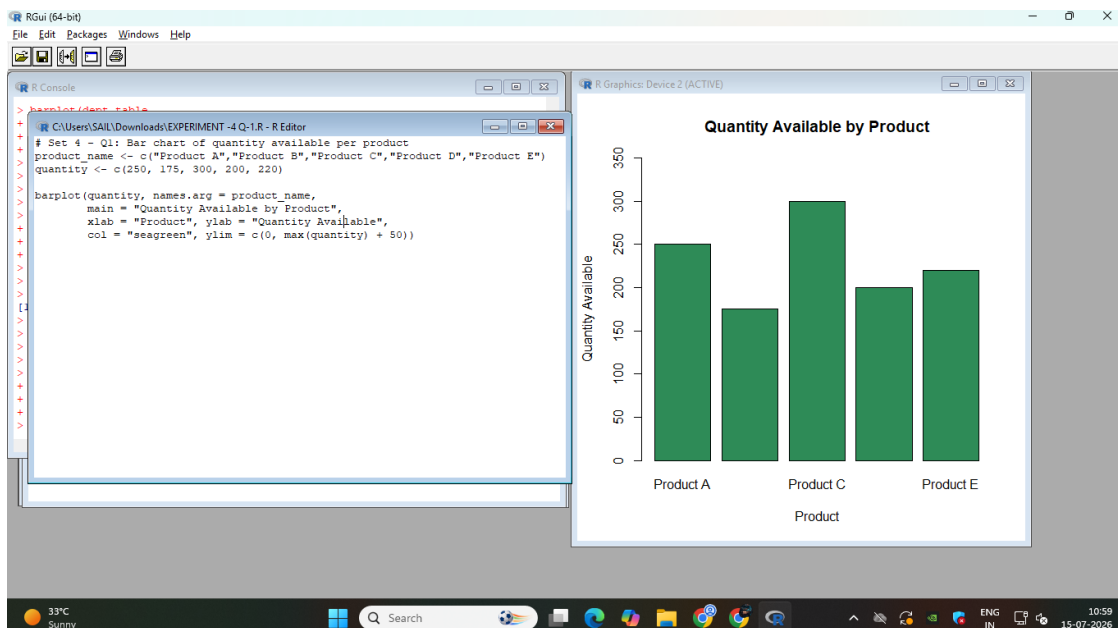
Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R Code

```
# Set 4 - Q1: Bar chart of quantity available per product
product_name <- c("Product A","Product B","Product C","Product D","Product E")
quantity <- c(250, 175, 300, 200, 220)

barplot(quantity, names.arg = product_name,
        main = "Quantity Available by Product",
        xlab = "Product", ylab = "Quantity Available",
        col = "seagreen", ylim = c(0, max(quantity) + 50))
```

Output



Question 2

Generate a stacked bar chart to show the quantity of each product within different product categories.

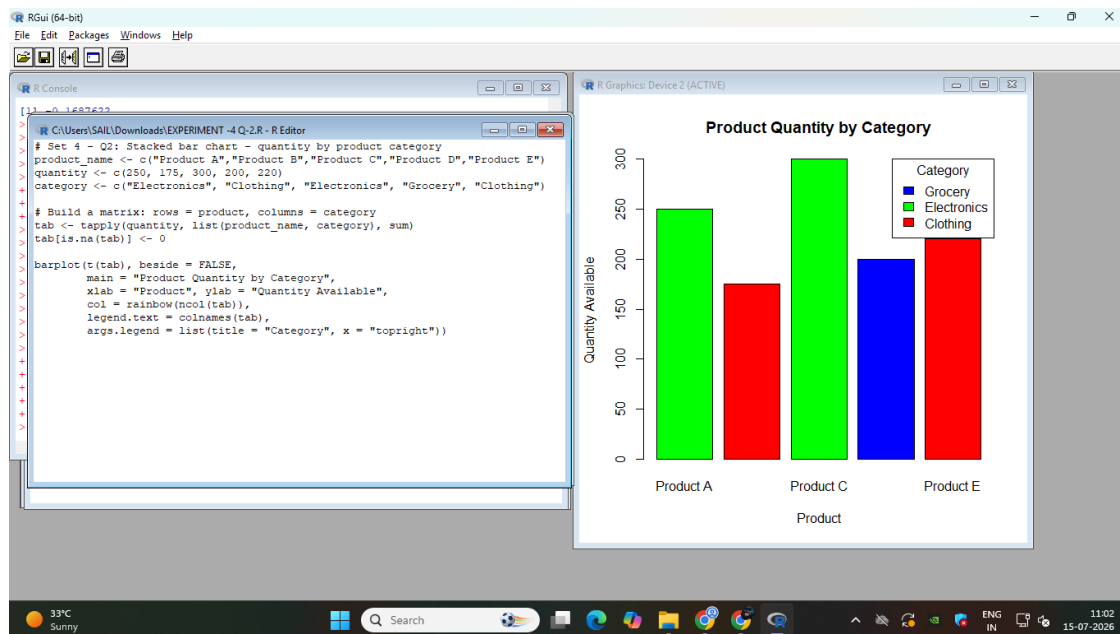
R Code

```
# Set 4 - Q2: Stacked bar chart - quantity by product category
product_name <- c("Product A","Product B","Product C","Product D","Product E")
quantity <- c(250, 175, 300, 200, 220)
category <- c("Electronics", "Clothing", "Electronics", "Grocery", "Clothing")

# Build a matrix: rows = product, columns = category
tab <- tapply(quantity, list(product_name, category), sum)
tab[is.na(tab)] <- 0

barplot(t(tab), beside = FALSE,
        main = "Product Quantity by Category",
        xlab = "Product", ylab = "Quantity Available",
        col = rainbow(ncol(tab)),
        legend.text = colnames(tab),
        args.legend = list(title = "Category", x = "topright"))
```

Output



Question 3

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

R Code

```
# Set 4 - Q3: Scatter plot - product price vs quantity available
product_name <- c("Product A","Product B","Product C","Product D","Product E")
quantity <- c(250, 175, 300, 200, 220)
price <- c(20, 35, 15, 28, 22) # sample data ($)
```

```

plot(price, quantity,
      main = "Product Price vs Quantity Available",
      xlab = "Price (in $)", ylab = "Quantity Available",
      pch = 19, col = "brown")
abline(lm(quantity ~ price), col = "blue", lwd = 2)

# Calculate and print the correlation
correlation <- cor(price, quantity)
cat("Correlation between Price and Quantity Available:", round(correlation, 3), "\n")

# Auto-generated finding based on the correlation value
if (correlation < -0.5) {
  cat("Finding: There is a strong negative correlation between price and quantity
  available.\n")
  cat("This suggests that lower-priced products tend to be stocked in higher
  quantities,\n")
  cat("possibly because they have higher turnover or are prioritized for bulk
  inventory.\n")
} else if (correlation > 0.5) {
  cat("Finding: There is a strong positive correlation between price and quantity
  available.\n")
  cat("This suggests that higher-priced products are being stocked in larger
  quantities.\n")
} else {
  cat("Finding: There is a weak or negligible correlation between price and quantity
  available.\n")
  cat("This suggests that price alone does not strongly determine how much of a product is
  stocked.\n")
}

```

Output

